

HARMONIC DRIVE

	Standard Gear	Space Specification Gear
Circular Spline	FCD 80 Ductile Iron	SUS 630 Stainless Steel
Flexspline	SNM 439 Alloy Steel	SUS 304L or 15-5PH Stainless Steel
Wave Bearing	SUJ 2 Bearing Steel	SUS 440C Stainless Steel
Wave Plug	S 45 C Carbon Steel	SUS 630 or SUS 304L Stainless Steel
Bearing Retainer	Nylon 66 with glass-fibre reinforcement	Phenolic Resin

SIEMENS

cmti

LINEAR MOTOR

Three-Phase Linear Motors 1FN3

Product Brief

03/2000

High
self-
acceleration

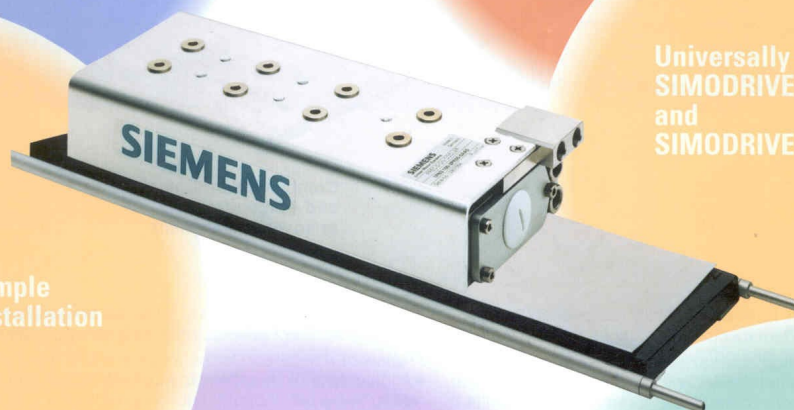
Optionally with
precise
thermal insulation

Universally adaptable to
SIMODRIVE 611 digital
and
SIMODRIVE 611 universal

Simple
installation

Extreme dynamic
performance and high
traversing rate

The new generation
of linear motors for
universal applications in
mechanical engineering

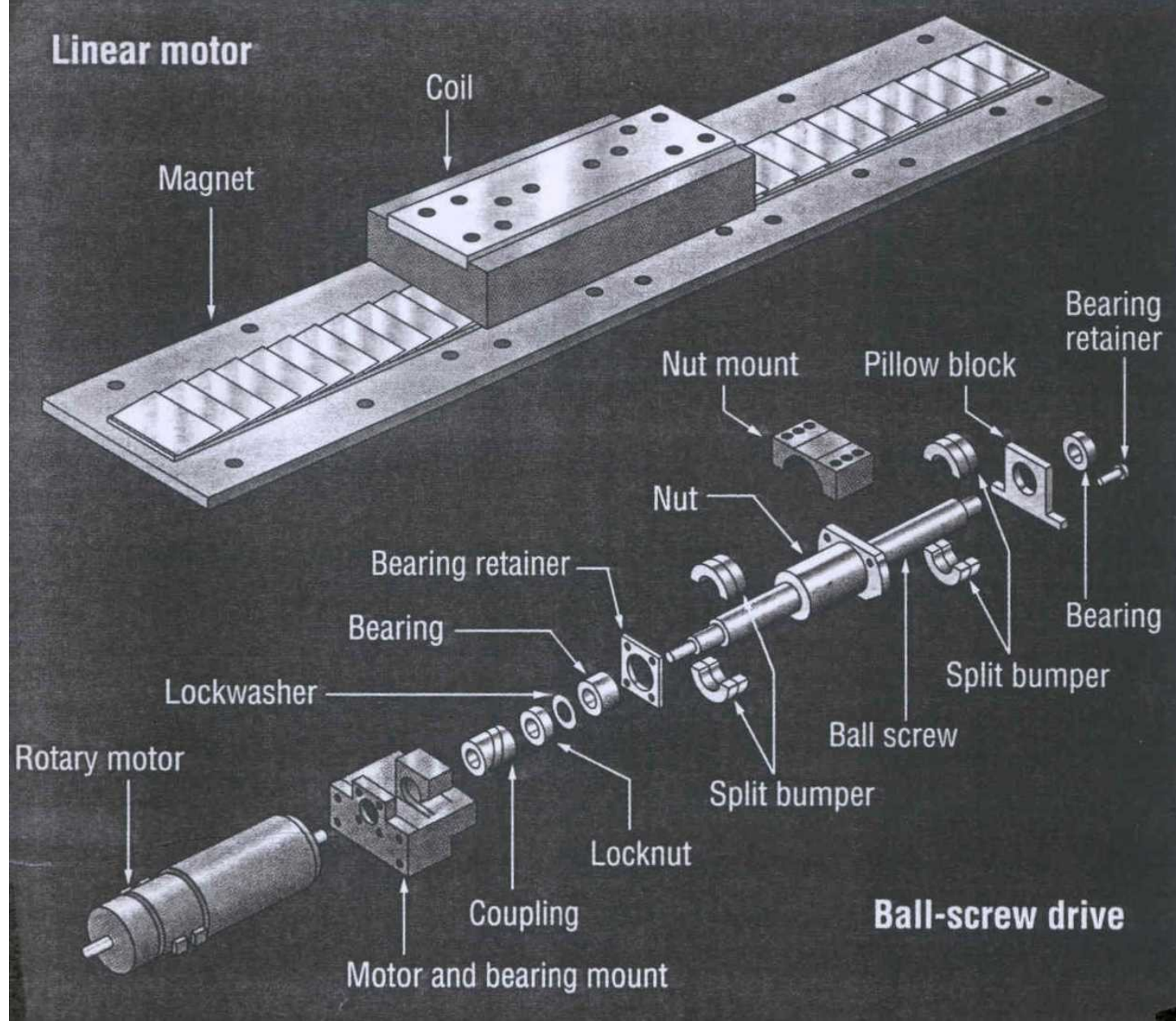


LINEAR MOTOR

- ❑ USER DEMAND FOR HIGH SPEED AND FEED IS INCREASING
- ❑ LINEAR MOTORS ARE BEING USED TO MEET HIGH SPEED AND HIGH FEED REQUIREMENTS
- ❑ LINEAR MOTOR IS SIMILAR TO A DEVELOPED ROTARY MOTOR
- ❑ THE SAME ELECTROMAGNETIC FORCES THAT PRODUCE TORQUE IN A ROTARY MOTOR ARE USED TO PRODUCE LINEAR FORCE IN LINEAR MOTORS

LINEAR MOTORS (Contd...)

- ❑ PROVIDE DIRECT LINEAR MOTION TO THE MOVING ELEMENT.
- ❑ THESE MOTORS ELIMINATE ALL OTHER DRIVING ELEMENTS LIKE BALL SCREW, COUPLING AND FEED MOTOR
- ❑ DUE TO THE ABSENCE OF THESE INTERMEDIATE PARTS, LINEAR MOTORS OFFER VERY HIGH STIFFNESS AND ACCURACY
- ❑ PROVIDE ACCURATE AND SMOOTH MOTION FREE FROM BACKLASH TO THE SLIDE
- ❑ TRAVEL IS UNLIMITED. NO WEAR OF PARTS



LINEAR MOTOR V/S BALL SCREW DRIVE

Performance comparison between linear motors and ball-screw drives

	Linear motors	Ball screw
Travel	Unlimited (standard)	5 m (special)
Velocity	10 m/sec (possible higher)	0.5 m/sec (critical speed)
Acceleration	10 g (high-force/inertia ratio)	5 g (high-reflected inertia)
Stiffness	1 000 000 N/mm (rigid structure)	(preload and friction dependent)
Accuracy	Submicron (sensor dependent)	Microns (friction limits repeatability)
Constant velocity	<1% (with air bearing)	>10% (jitters due to balls motion)

LIMITATIONS OF LINEAR MOTORS



1. The structural resonance frequency should be as high as possible typically 200 to 300 Hz.
2. The bearings should be selected to accommodate high magnetic attraction of the magnet assembly of linear motor
3. The high magnetic force exerts a heavy force on the guideways thus affecting its life.
4. Feed back devices should be mounted rigidly and seals should be designed for fluids, seals and chips.
5. Special linear encoders are required to work with high speeds.
6. Linear motors has put strong challenge to the manufacturers of high speed spindles